

Payments (Chapa)



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Single gateway integration for all paid actions in the app - subscriptions, promotions, reveals

Category: **Monetization** | Gateway: **Chapa (ETB)** | Status: **Production**

ETB Currency	Telebirr / CBE / card Methods	4 Payment types	Webhook + tx_ref Reconcile
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1. What it is

WaveMart's payment service is a thin, opinionated wrapper around Chapa, the dominant Ethiopian payment gateway. All monetary transactions on the platform - subscriptions, featured listings, VIP promotion, contact reveal - flow through this single, well-typed API.

2. How it works

01 Initialize 1	POST /api/payments/initialize { payment_type, amount, related_id } -> checkout_url, tx_ref, payment row.
02 WebView checkout 2	App opens Chapa checkout (WebView); user pays with Telebirr / CBE Birr / card.
03 Callback 3	Chapa POSTs /api/payment/callback (CSRF-exempt, no auth) with tx_ref and status.
04 Verify 4	GET /api/payments/verify/{tx_ref} re-verifies with Chapa server-side.
05 Activate entitlement 5	The same callback flips the user's subscription / listing / VIP flags.

3. Capabilities

Capability	What it does
Multi-type payments	Single endpoint handles subscription, featured_listing, vip_listing, contact_reveal.
Idempotent	tx_ref is the source of truth; replays are safe.
Webhook-driven	Chapa posts to /api/payment/callback with HMAC-equivalent verification.

History	GET /api/payments paginates the user's transactions with status.
Detail view	GET /api/payments/{id} shows one transaction in full.
Verification	GET /api/payments/verify/{tx_ref} re-checks status with Chapa.
Local UI	Payment history screen and detail screen built in Flutter.

4. Investor relevance

WHY IT MATTERS

Chapa is the only widely accepted online gateway in Ethiopia. By treating it as a pluggable service, WaveMart can additionally support direct bank integrations (CBE Birr, Awash, Dashen) or even crypto rails without touching feature code.

5. API

```
POST /api/payments/initialize { payment_type, amount, related_id }
GET /api/payments paginated history
GET /api/payments/{id} single transaction
GET /api/payments/verify/{tx_ref} re-verify with gateway
POST /api/payment/callback Chapa webhook (no auth, CSRF-exempt)
```